

Decision Analytics for Consulting

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Executive Summary

Aalto University's President's Management Team is reviewing the university's 10-year strategy for teaching and learning under uncertainty. To support this process, we developed a scenario-based decision support model that evaluates strategic action proposals across three alternative futures: Competitive Global Campus, Always-Learning Society and Human-First University.

Key scenarios

The model considers three future scenarios: Competitive Global Campus, Always-Learning Society and Human-First University. Using the Rank-Order Centroid (ROC) formula, the scenario probabilities were calculated based on plausibility ranking. Always-Learning Society was treated as the most likely scenario, while Competitive Global Campus was considered the least likely.

Objectives and attributes

The main objective of the model is to strengthen Aalto's position as a leading, future-ready university for learning and impact. Seven strategic action proposals were assessed using eight evaluation attributes covering learning outcomes, progression, learner experience, accessibility, assessment integrity, strategic autonomy, employability, and reputation. An additive value model, value functions, and swing weights were used to compare the actions consistently across scenarios.

Action proposals

Seven strategic action proposals were assessed: AI-Enabled Student Guidance, Elite Mentor Network / Smart Work-Study, AI Governance Board / Assessment Integrity, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials, Pedagogical Training for Staff, and Nordic-Baltic Tech Alliance. Each action was evaluated across all three scenarios using the selected attributes.

Portfolio model and recommendation

The portfolio model maximized expected portfolio utility under budget, time, implementation capacity, portfolio size, dependency, and balance constraints. The recommended base-case portfolio includes Elite Mentor Network / Smart Work-Study, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials, and Pedagogical Training for Staff.

Introduction

Aalto University's President's Management Team is reviewing the university's 10-year strategy for teaching and learning under uncertainty. Higher education is changing through innovations such as artificial intelligence, flexible learning needs, international competition, and changing values from students, employers, and society. These details make long-term strategic planning more complex since actions that perform well in one case may not be effective in another.

The purpose of this project is to support Aalto's strategy work by developing a scenario-based decision support model. The model evaluates strategic action proposals across three alternative futures: Competitive Global Campus, Always-Learning Society and Human-First University. These scenarios provide a structured way to analyze uncertainty and assess how different actions might contribute to achieving Aalto's long-term goals.

In this project, we acted both as consultants and as representatives of the client. We simulated the decision process by defining objectives, assessing actions, eliciting utility functions, weights and building a portfolio model according to the methods discussed in the course. This provides a transparent and structured basis for recommending actions that can help Aalto remain future-ready across uncertain conditions.

Scenarios

The decision model was developed using three alternative future scenarios: Competitive Global Campus, Always-Learning Society and Human-First University. They were used to represent uncertainty about the external environment in which Aalto's teaching and learning strategy may develop over the next decade. Instead of assuming one fixed future, the model evaluates each strategic action across all three scenarios.

Scenario 1: Competitive Global Campus

In this scenario, higher education becomes more selective and internationally competitive. Finnish universities are fewer but stronger, better funded, and more globally visible. Teaching emphasizes academic rigor, campus-based learning and human-led assessment. AI is used to support learning, but universities must manage risks related to fairness, academic integrity and dependence on external providers.

Scenario 2: Always-Learning Society

In this case, higher education shifts its focus toward lifelong learning and modular study paths. While AI supports personalization and guidance, learners combine education with work and other life situations. The main opportunities are flexibility and broader access, while the main risks include algorithmic overload and a less personal learning experience.

Scenario 3: Human-First University

In this scenario, universities focus strongly on resilience, trust and human-centered learning. AI is used more selectively and teaching emphasizes mentorship and presence-based pedagogy. Oral exams, project work and human judgment are the main assessments. The challenge here is balancing human-centered education with workload, scalability and innovation.

Elicitation of scenario probabilities

Scenario probabilities were elicited using the Rank-Order Centroid (ROC) method. First, the scenarios were ranked according to their perceived plausibility based on simulated management team discussion. The group ranked Always-Learning Society as the most likely scenario, followed by Human-First University and then Competitive Global Campus.

The ROC method was used because it provides a structured way to convert ordinal rankings into probabilities without requiring precise numerical judgments. This was suitable for the project since the probabilities are judgment-based and future-oriented. Also, it helps with the biases mitigation such as overconfidence and anchoring, by providing a structured approach to weighting scenarios.

Scenario	Probability
Competitive Global Campus	0.111
Always-Learning Society	0.611
Human-First University	0.278

Table 1: Scenario Probabilities

These probabilities were then used in the model to calculate the expected utility of each strategic action across the three scenarios. Since scenario probabilities are uncertain, their effect on the final recommendation was later tested in the sensitivity analysis.

The overall objective is to strengthen Aalto's position as a leading, future-ready university for learning and impact. Following a value-focused framing, the analysis decomposes this objective into five main objective areas: learning effectiveness, learner experience, academic excellence, institutional strength and societal impact. This structure helped distinguish the ends Aalto seeks to achieve from the actions used to achieve them (Keeney, 1994).

The value tree illustrates how the overall objective is decomposed into objective areas and attributes. Learning effectiveness is measured through achievement of learning outcomes and completion/progression. Learner experience is measured through engagement and accessibility/flexibility/equity. Academic excellence is represented by integrity of assessment. Institutional strength is captured through strategic control/autonomy and global attractiveness/reputation. Societal impact is measured through external relevance and employability. In line with value-focused thinking, the tree clarifies the valued ends before detailed action evaluation begins (Keeney, 1994).

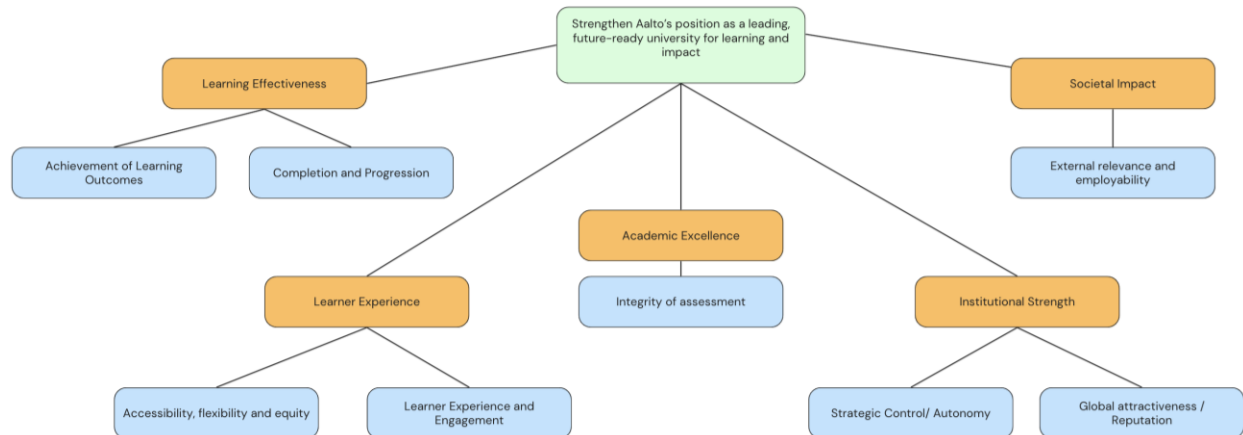


Figure 1. Value Tree

Attribute	Description
Achievement of learning outcomes	Measures expected effect on student competency development
Completion and progression	Captures impact on study progress and completion
Learner experience and engagement	Assesses student engagement and learning experience
Accessibility, flexibility and equity	Evaluates support for different learner needs and life situations
Integrity of assessment	Measures trustworthiness and credibility of assessment
Strategic control / autonomy	Captures Aalto's control over key learning infrastructure and decisions
External relevance and employability	Evaluates connection to working life and graduate employability
Global attractiveness / reputation	Measures contribution to Aalto's international visibility

Table 2: Attribute explanation

To evaluate the action proposals consistently, each attribute was assigned a measurement scale with a worst and best plausible consequence. Some attributes used natural percentage scales, such as achievement of learning outcomes, completion and progression, strategic

control/autonomy, and external relevance/employability. Other attributes used constructed categorical scales, such as learner experience and engagement, integrity of assessment, and global attractiveness/reputation. Accessibility, flexibility and equity was measured as a count scale from 0 to 4, representing how many learner profiles are supported. These scales helped to include both quantitative and qualitative impacts in the evaluation model.

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The strategic action proposals were defined through group discussion using the three scenarios and the value tree. The goal was to identify actions that could help Aalto perform well across different possible futures of higher education. The group considered actions that would support learning effectiveness, learner experience, academic excellence, institutional strength and societal impact. The process followed the logic of value-focused brainstorming, in which explicit objectives are used to stimulate the creation of higher-value alternatives rather than simply screening an initial list of familiar options (Keeney, 2012).

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The final list includes seven action proposals:

Action proposal	Description
AI-Enabled Student Guidance	Uses AI-supported tools to provide personalized academic guidance and early support for students
Elite Mentor Network / Smart Work-Study	Connects students with mentors, employers and work-study opportunities
AI Governance Board / Assessment Integrity	Creates governance principles for responsible AI use and trustworthy assessment
Aalto-Controlled Core Infrastructure	Develops secure internal digital infrastructure for learning and data control

Aalto-Certified Micro-Credentials	Offers modular certified learning units for flexible and lifelong learning
Pedagogical Training for Staff	Trains staff to adapt teaching, assessment and guidance to future learning needs
Nordic-Baltic Tech Alliance	Builds regional partnerships to strengthen Aalto's external relevance and reputation

Table 3: Action proposals

The proposals were evaluated across all three scenarios, and within each scenario, every action was assessed using the eight attributes defined in the value tree. The scenario-specific performance table illustrates how the value of each action changes depending on the future context. For example, AI-Enabled Student Guidance performs strongly in the Always-Learning Society scenario because personalized guidance and continuous support are more important in a lifelong learning environment. In contrast, Aalto-Controlled Core Infrastructure is particularly valuable in scenarios where strategic autonomy, data control, and reliable digital infrastructure are essential. These performance assessments were then used as inputs for the utility functions, attribute weights and portfolio optimization model.

Elicitation of utility functions

The action proposals were assessed on eight attributes that use different original scales: percentages for learning outcomes, completion, strategic control, and employability; count scales for accessibility; and anchored qualitative categories for learner experience, assessment integrity, and global attractiveness. To make these heterogeneous measures comparable, the team converted raw performances into attribute-specific utility (value) scores on a common 0-1 scale. This explicit transformation helps avoid the common mistake of comparing unlike scales directly when making value trade-offs (Keeney, 2002).

Verification of the underlying assumptions

Before applying an additive model, the team checked whether it was reasonable to combine attribute-specific scores into a single overall value. In practice, this was done by examining whether preferences over one attribute or a small set of attributes remained stable when the levels of the remaining attributes were held constant at different levels. These checks were used to assess whether an additive value model was a sufficiently good representation of the management team's judgments. The conclusion was that the assumptions were supported well enough for practical decision support, which justified the later use of weighted summation. This is consistent with the additive portfolio utility logic used in portfolio decision analysis when attributes can be

treated as sufficiently separable for practical modelling (Vilkkumaa, Rikkinen and Liesiö, 2025).

This verification step is important for the client because the final recommendations should not rest on a mechanical formula alone. The additive form is only defensible if the preference structure is stable enough that the value of an improvement on one attribute can be assessed separately and then combined with the others.

Elicitation process

Two elicitation methods were used. For the natural percentage attributes, the team applied the bisection method. Each such attribute was anchored at a worst and a best plausible consequence, after which midpoint, quarter-point, and three-quarter-point levels were elicited. Intermediate values were then calculated by piecewise linear interpolation. For the qualitative and count-based attributes, the team used direct rating. The worst level was assigned value 0, the best level value 1, and the intermediate levels were rated directly. This hybrid approach supports clear trade-off judgments while reducing the risk of poorly grounded or inconsistent swaps between unlike consequences (Keeney, 2002).

This hybrid approach was chosen because it balances rigor with practicality. It keeps the questions intuitive for the decision makers, reduces the cognitive burden associated with lottery-based questioning, and still produces explicit value functions that can be audited, illustrated, and stress-tested. For a client-facing strategic process, this transparency matters because the resulting recommendations should be understandable, discussable, and revisable rather than dependent on opaque scoring rules (Keeney, 2002).

$$U_{a,s} = \sum_{i \in I} w_i u_i(x_{a,i,s})$$

Equation 1. Scenario-specific value of action a, obtained by summing weighted attribute-specific values.

$$EU_a = \sum_{s \in S} p_s U_{a,s}$$

Equation 2. Expected utility of action a, obtained by combining scenario-specific values with scenario probabilities.

In Equation 1, $x_{a,i,s}$ denotes the raw performance of action a on attribute i in scenario s, $u_i(\cdot)$ is the attribute-specific utility function, and w_i is the normalized weight of attribute i. Equation 2 then combines the scenario-specific values using the elicited scenario

probabilities. This is the bridge between the raw performance tables in the workbook and the action-level expected utilities used in the portfolio model.

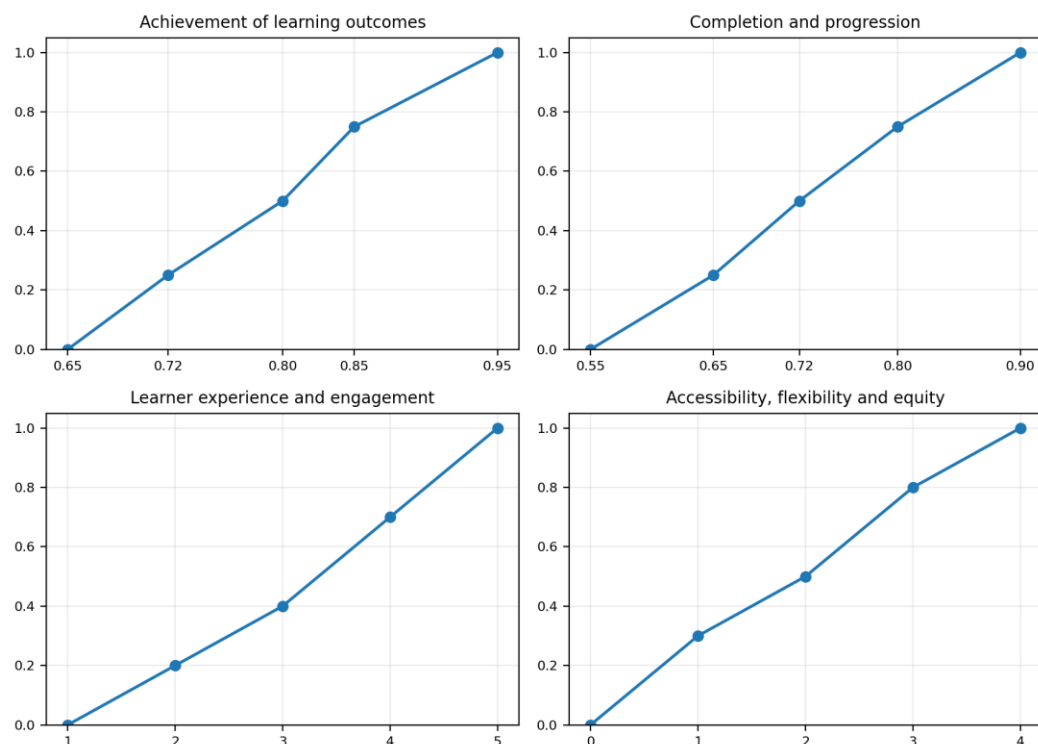


Figure 2. Utility functions for learning outcomes, completion and progression, learner experience, and accessibility.

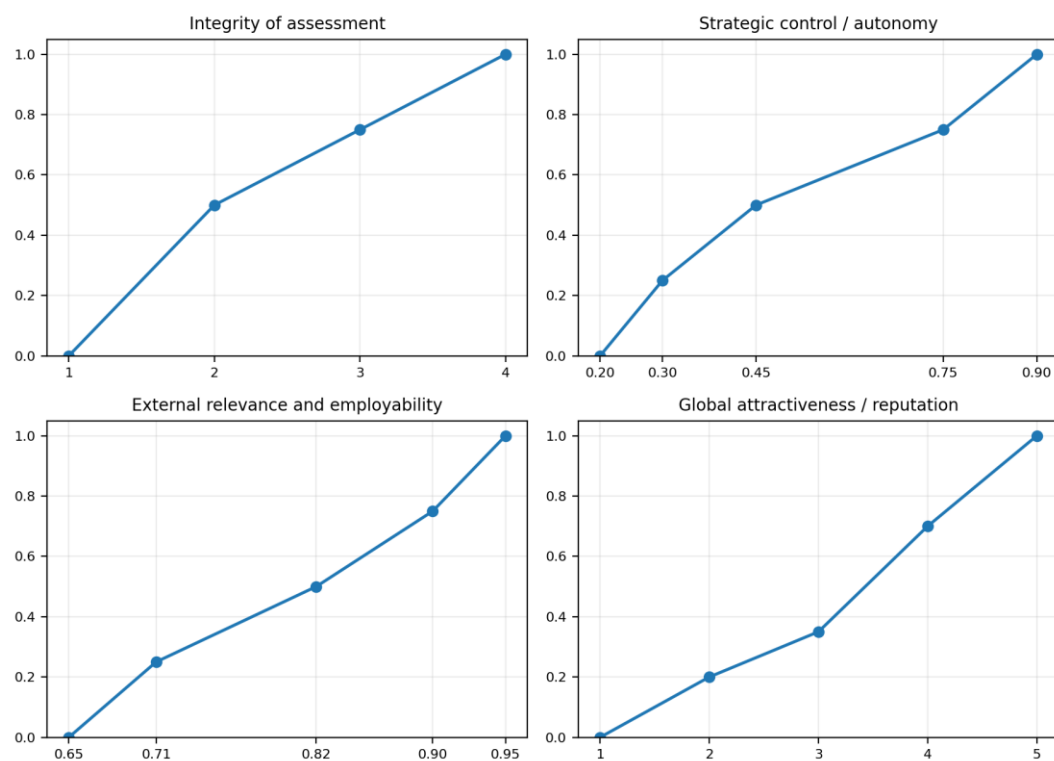


Figure 3. Utility functions for assessment integrity, strategic control, employability, and global attractiveness.

The figures show that the team did not assume all attributes to be linear. Some attributes were treated as approximately linear, where value increases steadily across the range, while others were intentionally more concave or convex. For example, the utility function for assessment integrity places a high premium on moving away from a fragile assessment regime, while the functions for external relevance and global attractiveness become steeper toward the upper end of the scale. This allows the model to reflect the fact that some improvements matter more than others, even when the raw change on the original scale is the same.

Elicitation of attribute weights / scaling constants

After the utility functions had been established, the next task was to determine how much each attribute should contribute to the overall assessment. The team used a swing-based ranking exercise anchored on an all-worst benchmark. The benchmark consequence fixed all eight attributes at their worst plausible levels. From there, each attribute was considered one at a time by comparing the consequence of improving only that attribute from its worst to its best level while leaving all other attributes unchanged. This structure is important because useful value trade-offs depend on explicit consequence ranges rather than abstract statements of importance (Keeney, 2002).

These worst-to-best swings were ranked from the most valuable to the least valuable. The resulting rank order was then converted into final normalized weights using a rank-order centroid (ROC) rule. This gave a transparent and operational weighting procedure while keeping the elicitation manageable in a group setting. Using a structured weighting rule also reduces the temptation to overstate precision in judgments that are inherently comparative and qualitative (Keeney, 2002).

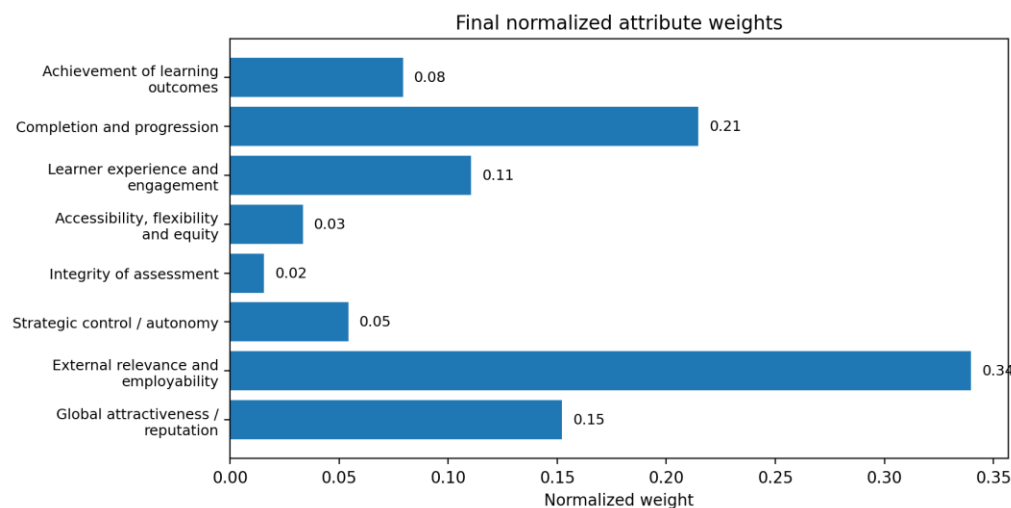


Figure 4. Final normalized attribute weights after ranking worst-to-best swings and converting them with ROC.

The resulting weights reveal a clear emphasis on external relevance and employability, completion and progression, and global attractiveness. In the final model, external relevance and employability receives the largest weight (0.340), followed by completion and progression (0.215) and global attractiveness (0.152). Learning outcomes and learner experience remain important, but they carry smaller weights in the final aggregation. Accessibility, strategic control, and especially integrity of assessment receive lower weights in this particular weighting round.

Attribute	Final normalized weight	Interpretation
External relevance and employability	0.340	Largest swing; strong emphasis on labour-market relevance and visible stakeholder value
Completion and progression	0.215	Strong emphasis on student success through to completion
Global attractiveness / reputation	0.152	Reputation matters for long-run positioning and talent attraction
Learner experience and engagement	0.111	Important, but not the dominant driver of the final recommendation
Achievement of learning outcomes	0.079	Valued, but given less incremental weight than progression and employability

For the client, the managerial interpretation of these weights is more important than the exact decimals. The model gives priority to actions that help Aalto remain relevant to working life, improve the probability of successful progression, and protect or strengthen its long-term attractiveness. This explains why employability-oriented and pathway-oriented actions perform strongly in the final portfolio.

Portfolio model

The portfolio model translates the preceding preference modeling into a concrete resource-allocation problem. Each strategic action is assigned an expected utility, and the model selects the combination of actions that maximizes the total expected portfolio utility subject to practical implementation constraints. This is consistent with the logic of resource-allocation decision analysis, in which the organization seeks the highest total benefit from a limited resource envelope rather than evaluating projects in isolation (Kleinmuntz, 2007).

$$\max \sum_{a \in A} z_a EU_a$$

Equation 3. Portfolio objective: select the set of actions that maximizes total expected utility.

In Equation 3, z_a is a binary decision variable that equals 1 when action a is selected and 0 otherwise. EU_a denotes the expected utility of action a , computed from Equations 1 and 2. The model therefore uses the value functions, attribute weights, and scenario probabilities as inputs, but the optimization itself is performed at the action level. This formulation is also aligned with recent scenario-based action-portfolio models that combine scenario-contingent impacts, preference information, and feasibility constraints in one optimization framework (Vilkkumaa, Rikkinen and Liesiö, 2025).

Mathematical formulation

- Decision variables: $z_a \in \{0,1\}$ for each of the seven strategic actions.
- Objective: maximize $\sum_a z_a EU_a$.
- Budget constraint: total cost of selected actions $\leq \text{€}5.0\text{M}$.
- Implementation-capacity constraint: total capacity points of selected actions ≤ 6 .
- Portfolio-size constraint: no more than 4 actions can be selected.
- Dependency constraint: Micro-Credentials can only be selected if Core Infrastructure is selected.
- Balance constraint: the portfolio must include at least one outward-facing action.
- Time constraint: total implementation time of selected actions ≤ 10 years.

In addition to budget and time, the portfolio model includes an implementation-capacity constraint. Capacity points were introduced to represent the internal organizational effort required to implement a strategic action, including managerial attention, cross-unit coordination, staff involvement, IT integration, governance effort, and change-management burden. This constraint was needed because financial affordability and long-term time feasibility alone do not capture whether Aalto can realistically implement several demanding initiatives in parallel. Thus, capacity points serve as a practical proxy for organizational bandwidth. Actions with broader coordination requirements or heavier implementation overhead were assigned higher capacity-point scores than lighter actions. This kind of additional nonfinancial resource constraint is a standard extension in portfolio decision analysis when scarce organizational bandwidth, expertise, or oversight capacity can become binding even when the monetary budget is not (Kleinmuntz, 2007).

The dependency between Micro-Credentials and Core Infrastructure reflects a practical implementation logic. Aalto-certified micro-credentials require a digital backbone for delivery, identity management, certification, and data handling. The balance requirement was introduced so that the portfolio would not become purely internal; at least one selected action must create visible stakeholder value for learners, employers, or partners.

Selected base-case action	Cost (M€)	Capacity points	Primary role
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Elite Mentor Network / Smart Work-Study	1.0	1	Employability and work-integrated learning
Aalto-Controlled Core Infrastructure	2.0	2	Digital autonomy and trusted delivery backbone
Aalto-Certified Micro-Credentials	1.1	1	Flexible pathways and lifelong learning
Pedagogical Training for Staff	0.8	2	Teaching capability and implementation readiness

Decision recommendations

The recommended portfolio was obtained in four steps. First, raw action performances were estimated for each attribute in each scenario. Second, those performances were converted into utility scores using the elicited utility functions. Third, the weighted scenario-specific utilities were combined into expected utilities using the scenario probabilities. Finally, the portfolio optimization model selected the feasible set of actions with the highest expected total utility. This sequence mirrors the standard progression from value modelling to constrained portfolio selection in decision analytics for resource allocation (Kleinmuntz, 2007; Vilkkumaa, Rikkinen and Liesiö, 2025).

Under the base-case assumptions, the model recommends a four-action portfolio consisting of Elite Mentor Network / Smart Work-Study, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials, and Pedagogical Training for Staff. The expected utility of this portfolio is 2.462, with a budget use of €4.9M, full use of the six implementation-capacity points, and a total implementation time of 5.2 years.

Selected action	Why the model selected it	Client-facing contribution	Role in the portfolio
Elite Mentor Network / Smart Work-Study	Consistently high value on employability and engagement; selected in every sensitivity case.	Strengthens the link between learning and working life; improves external relevance and learner motivation.	Most stable “no-regret” action; anchors the outward-facing value of the portfolio.
Aalto-Controlled Core Infrastructure	Needed to unlock the Micro-Credentials action and strengthens scenario performance where autonomy matters.	Reduces dependence on external providers and gives Aalto more control over key digital learning systems.	Foundational enabling action; supports resilience and trusted execution.

Aalto-Certified Micro-Credentials	High stand-alone expected utility and strong fit with the most likely scenario, Always-Learning Society.	Expands modular learning, lifelong learning, and reskilling opportunities for different learner groups.	Flexible pathway action; strengthens adaptability and future relevance.
Pedagogical Training for Staff	Selected because it improves the ability to implement future teaching and assessment models across scenarios.	Builds staff capability, teaching quality, and practical implementation readiness.	Capability-building action; raises the effectiveness of the broader portfolio.

The recommendation should be understood as a complementary package rather than as four independent projects. The portfolio combines outward-facing value for learners and employers with institution-building actions that make new delivery models credible and scalable. It also explains why some actions with attractive stand-alone utility, such as AI-Enabled Student Guidance, are not selected in the base case: once dependencies and resource constraints are taken into account, the selected combination creates greater total portfolio value.

Sensitivity analyses

Sensitivity analysis was used to test whether the recommendation is robust to reasonable changes in the key assumptions. The team varied one block of inputs at a time and then re-solved the portfolio model. The tested cases included equal scenario probabilities, stronger weighting for Scenario 1, stronger weighting for Scenario 3, tighter and looser budget limits, tighter implementation capacity, and a reduced maximum number of selected actions. This kind of structured stress testing is especially important in scenario-based portfolio analysis because scenario probabilities and preference parameters are judgment-based and should therefore be tested rather than treated as immutable facts (Vilkkumaa, Rikkinen and Liesiö, 2025).

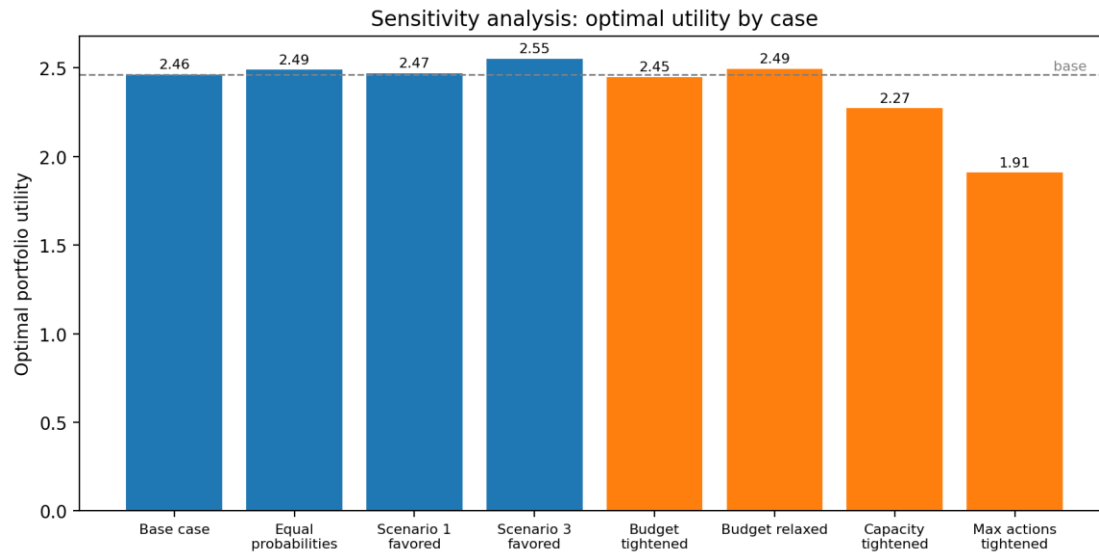


Figure 4. Optimal portfolio utility under the tested sensitivity cases.

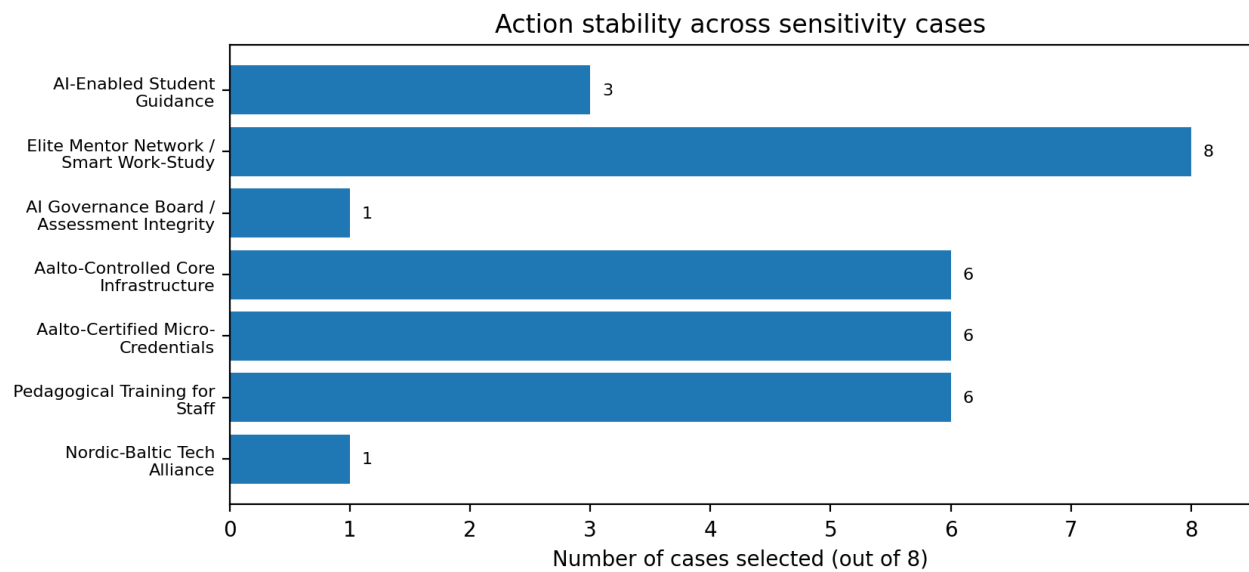


Figure 5. Selection frequency of each action across the eight sensitivity cases.

Sensitivity case	Optimal utility	Portfolio changed?	Interpretation
Equal probabilities	2.489	No	The base-case portfolio remains optimal even when the scenario ranking is ignored.

Scenario 1 favored	2.470	No	The base-case portfolio remains optimal under a more competitive-global view.
Scenario 3 favored	2.553	No	The base-case portfolio remains optimal and performs especially well under a stronger human-first view.
Budget tightened	2.449	Yes	The model replaces infrastructure-heavy elements with a lighter outward-facing package.
Budget relaxed	2.495	Yes	Additional resources allow AI-Enabled Student Guidance to enter the portfolio.
Capacity tightened	2.272	Yes	A tighter implementation bottleneck triggers a shift toward a lower-bandwidth governance mix.
Max actions tightened	1.909	Yes	The largest value loss occurs when the portfolio is forced to become too narrow.

Two findings are particularly important for the client. First, the recommendation is robust to changes in the scenario probabilities: the base-case portfolio remains optimal in every probability-based test. This indicates that the recommendation is not driven by one fragile view of the future. Second, the recommendation is more sensitive to implementation constraints. Tightening the budget, capacity limit, or maximum number of actions changes the optimal portfolio, and the strongest utility loss occurs when the maximum number of actions is reduced. Managerially, this means that Aalto's main vulnerability is not moderate disagreement about future scenario likelihoods, but the practical challenge of implementing enough complementary actions at the same time.

Action stability provides an additional robustness insight. Elite Mentor Network / Smart Work-Study is selected in all eight cases, making it the most stable and most clearly 'no-regret' action in the model. Core Infrastructure, Micro-Credentials, and Pedagogical Training are also selected frequently, which confirms their central role in the overall strategy. By contrast, AI Governance and Nordic-Baltic Tech Alliance enter only in more specialised constrained cases.

Behavioral and group aspects

Because this project simulated a management-team decision process, it was important to recognize that the results are shaped not only by data and model structure, but also

by human judgment. In this kind of setting, several cognitive biases can affect the process: anchoring on early numbers or rankings, overconfidence in scenario probabilities or performance estimates, desirability bias toward favored actions, and range insensitivity when judging swings across attributes. These risks are well aligned with the broader literature on value trade-offs and structured judgment (Keeney, 2002).

Group dynamics create a second source of risk. Stronger personalities may influence rankings and assessments more than others, while quick agreement can lead to groupthink rather than careful challenge of assumptions. Unequal participation is also a concern, especially in tasks such as scenario assessment, value-function elicitation, and weighting, where some members may feel less confident contributing. As a result, there is a risk that the final model reflects facilitation choices and discussion dynamics as much as the true preferences of the decision makers. This is especially relevant because the model depends on judgment-based inputs such as scenario probabilities, attribute weights, value-function shapes, and action performance estimates.

To reduce these risks, the project used a structured facilitation approach. Scenario probabilities were elicited through a Rank-Order Centroid (ROC) procedure rather than requiring precise probability statements from the outset. Attribute weights were elicited through worst-to-best swing comparisons, and value functions used common anchors and standardized shapes where possible. The group also split contributions across scenarios and model components, reducing the chance that a single person's framing would dominate the whole analysis. This is broadly consistent with value-focused approaches that make objectives explicit before generating and evaluating alternatives, and with portfolio frameworks that encourage transparent and challengeable strategic discussion (Keeney, 1994; Keeney, 2012; Vilkkumaa, Rikkinen and Liesiö, 2025).

The group process also benefited from the fact that the model provided a shared structure for discussion. Instead of debating actions only at the level of general opinion, the team compared proposals across clearly defined attributes, scenarios, and implementation constraints. This helped separate personal preference from portfolio logic. Sensitivity analysis further strengthened the process by testing whether the recommendation changed when key assumptions were varied. In this way, the model served not just as a calculation tool, but as a disciplined framework for collective reasoning under uncertainty.

Conclusion

This report developed a scenario-based decision support model to evaluate strategic teaching-and-learning actions for Aalto University under conditions of long-term uncertainty. By combining scenario analysis, value-focused structuring, explicit utility functions, swing-based weighting, and constrained portfolio optimization, the model

provided a transparent basis for choosing a robust set of strategic actions (Keeney, 1994; Kleinmuntz, 2007; Vilkkumaa, Rikkinen and Liesiö, 2025).

Under the base-case assumptions, the recommended portfolio consists of four complementary actions: **Elite Mentor Network / Smart Work-Study, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials, and Pedagogical Training for Staff**. Together, these actions form a balanced package that strengthens both learner-facing value and institutional capability. The portfolio improves employability and work-integrated learning, supports modular and lifelong learning pathways, reinforces Aalto's control over critical learning infrastructure, and builds staff readiness for future teaching and assessment models. In this sense, the recommendation is not a set of isolated projects, but an integrated strategic portfolio whose elements reinforce one another.

A central finding of the analysis is that the recommended portfolio is robust to changes in scenario probabilities. Across the probability-based sensitivity tests, the same portfolio remained optimal, suggesting that the recommendation does not depend on one highly specific view of the future. At the same time, the results show that the recommendation is more sensitive to implementation constraints than to probability shifts. Changes in budget, implementation capacity, or portfolio-size limits alter the selected portfolio and reduce overall utility, with the strongest value loss occurring when the number of selectable actions is tightly constrained. This indicates that organizational bandwidth and the ability to maintain a sufficiently broad strategic portfolio are critical to successful implementation.

The analysis also identifies Elite Mentor Network / Smart Work-Study as the most robust individual action, as it is selected in every tested sensitivity case. Core Infrastructure, Micro-Credentials, and Pedagogical Training also appear frequently, confirming their importance within the broader strategy. Overall, the main strategic implication is that Aalto should prioritize a balanced portfolio of mutually reinforcing actions rather than rely on a single major initiative. Long-term success is likely to depend on strengthening both institutional foundations and learner-facing relevance at the same time, thereby enabling resilience across uncertain future conditions.

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Appendix

The appendix provides workbook-based supporting material for the main analysis, including scenario probabilities, attribute scales, raw action-performance assessments, attribute-specific value scores, portfolio input data, and the tested sensitivity cases.

Appendix A. Scenario probability elicitation

Scenario	Rank	Probability
Competitive Global Campus	3	11.1%
Always-Learning Society	1	61.1%
Human-First University	2	27.8%

Appendix B. Attribute scales and anchors

Attribute	Scale type	Worst plausible consequence	Best plausible consequence	Scale description
Achievement of learning outcomes	Natural percentage	65%	95%	% of students meeting competency targets
Completion and progression	Natural percentage	55%	90%	% of students completing within target time or target+1 year
Learner experience and engagement	Anchored categorical	Very poor	Excellent	1=Very poor; 2=Poor; 3=Acceptable; 4=Good; 5=Excellent
Accessibility flexibility and equity	Count / extent	0	4	Number of learner profiles well supported: 0=None; 1=One; 2=Two; 3=Three; 4=All four
Integrity of assessment	Anchored categorical	Fragile	Exemplary	1=Fragile; 2=Basic; 3=Robust; 4=Exemplary
Strategic control / autonomy	Natural percentage	20%	90%	% of critical learning/assessment/data infrastructure under Aalto or trusted regional control
External relevance and employability	Natural percentage	65%	95%	percentage of graduates in relevant employment or further study within 12 months
Global attractiveness / reputation	Anchored categorical	Local	Globally distinctive	1=Local; 2=National; 3=Nordic-Baltic; 4=European; 5=Globally distinctive

Appendix C. Swing ranking and normalized attribute weights

Attribute	Worst-to-best swing consequence	Rank	Final normalized weight
Achievement of learning outcomes	95%,55%,1,0,1,20%,65%,1	5	0.079
Completion and progression	65%,90%,1,0,1,20%,65%,1	2	0.215
Learner experience and engagement	65%,55%,5,0,1,20%,65%,1	4	0.111
Accessibility flexibility and equity	65%,55%,1,4,1,20%,65%,1	7	0.033
Integrity of assessment	65%,55%,1,0,4,20%,65%,1	8	0.016
Strategic control / autonomy	65%,55%,1,0,1,90%,65%,1	6	0.054
External relevance and employability	65%,55%,1,0,1,20%,95%,1	1	0.340
Global attractiveness / reputation	65%,55%,1,0,1,20%,65%,5	3	0.152

Appendix D.1. Raw action-performance assessments - Competitive Global Campus

Action	Achievement of learning outcomes (65-95%)	Completion and progression (55-90%)	Learner experience and engagement	Accessibility, flexibility and equity (0-4)	Integrity of assessment	Strategic control / autonomy (20-90%)	External relevance and employability (65-95%)	Global attractiveness / reputation (1-5)
AI-Enabled Student Guidance	82%	78%	Good	3	Basic	40%	82%	3
Elite Mentor Network / Smart Work-Study	85%	75%	Excellent	1	Robust	45%	90%	4
AI Governance Board / Assessment Integrity	75%	65%	Poor	2	Exemplary	60%	80%	3
Aalto-Controlled Core Infrastructure	80%	72%	Acceptable	2	Robust	80%	82%	2
Aalto-Certified Micro-Credentials	80%	75%	Good	4	Basic	50%	88%	4
Pedagogical Training for Staff	85%	78%	Good	2	Robust	50%	84%	3
Nordic-Baltic Tech Alliance	82%	75%	Acceptable	1	Robust	65%	86%	4

Appendix D.2. Raw action-performance assessments - Always-Learning Society

Action	Achievement of learning outcomes (65-95%)	Completion and progression (55-90%)	Learner experience and engagement	Accessibility, flexibility and equity (0-4)	Integrity of assessment	Strategic control / autonomy (20-90%)	External relevance and employability (65-95%)	Global attractiveness / reputation (1-5)
AI-Enabled Student Guidance	88%	85%	Acceptable	4	Fragile	35%	85%	3
Elite Mentor Network / Smart Work-Study	82%	80%	Excellent	1	Robust	40%	95%	4
AI Governance Board / Assessment Integrity	78%	68%	Poor	2	Exemplary	55%	82%	4
Aalto-Controlled Core Infrastructure	82%	75%	Poor	3	Robust	85%	85%	2
Aalto-Certified Micro-Credentials	85%	82%	Good	4	Basic	45%	92%	4
Pedagogical Training for Staff	88%	80%	Good	3	Robust	45%	85%	3
Nordic-Baltic Tech Alliance	80%	72%	Acceptable	2	Robust	55%	88%	3

Appendix D.3. Raw action-performance assessments - Human-First University

Action	Achievement of learning outcomes (65-95%)	Completion and progression (55-90%)	Learner experience and engagement	Accessibility, flexibility and equity (0-4)	Integrity of assessment	Strategic control / autonomy (20-90%)	External relevance and employability (65-95%)	Global attractiveness / reputation (1-5)
AI-Enabled Student Guidance	80%	75%	Very poor	2	Fragile	45%	80%	2
Elite Mentor Network / Smart Work-Study	80%	72%	Good	0	Robust	50%	88%	3
AI Governance Board / Assessment Integrity	75%	65%	Acceptable	1	Exemplary	75%	78%	4

Aalto-Controlled Core Infrastructure	78%	70%	Acceptable	1	Exemplary	90%	80%	3
Aalto-Certified Micro-Credentials	75%	70%	Poor	2	Basic	55%	85%	2
Pedagogical Training for Staff	82%	75%	Acceptable	1	Robust	55%	82%	3
Nordic-Baltic Tech Alliance	85%	78%	Good	1	Exemplary	85%	92%	5

Appendix E.1. Attribute-specific value scores - Competitive Global Campus

Action	Achievement of learning outcomes (65-95%)	Completion and progression (55-90%)	Learner experience and engagement	Accessibility, flexibility and equity (0-4)	Integrity of assessment	Strategic control / autonomy (20-90%)	External relevance and employability (65-95%)
AI-Enabled Student Guidance	0.600	0.688	0.700	0.800	0.500	0.417	0.500
Elite Mentor Network / Smart Work-Study	0.750	0.594	1.000	0.300	0.750	0.500	0.750
AI Governance Board / Assessment Integrity	0.344	0.250	0.200	0.500	1.000	0.625	0.455
Aalto-Controlled Core Infrastructure	0.500	0.500	0.400	0.500	0.750	0.833	0.500
Aalto-Certified Micro-Credentials	0.500	0.594	0.700	1.000	0.500	0.542	0.688
Pedagogical Training for Staff	0.750	0.688	0.700	0.500	0.750	0.542	0.562
Nordic-Baltic Tech Alliance	0.600	0.594	0.400	0.300	0.750	0.667	0.625

Appendix E.2. Attribute-specific value scores - Always-Learning Society

Action	Achievement of learning outcomes (65-95%)	Completion and progression (55-90%)	Learner experience and engagement	Accessibility, flexibility and equity (0-4)	Integrity of assessment	Strategic control / autonomy (20-90%)	External relevance and employability (65-95%)
AI-Enabled Student Guidance	0.825	0.875	0.400	1.000	0.000	0.333	0.594
Elite Mentor Network / Smart Work-Study	0.600	0.750	1.000	0.300	0.750	0.417	1.000
AI Governance Board / Assessment Integrity	0.438	0.357	0.200	0.500	1.000	0.583	0.500
Aalto-Controlled Core Infrastructure	0.600	0.594	0.200	0.800	0.750	0.917	0.594
Aalto-Certified Micro-Credentials	0.750	0.800	0.700	1.000	0.500	0.500	0.850
Pedagogical Training for Staff	0.825	0.750	0.700	0.800	0.750	0.500	0.594
Nordic-Baltic Tech Alliance	0.500	0.500	0.400	0.500	0.750	0.583	0.688

Appendix E.3. Attribute-specific value scores - Human-First University

Action	Achievement of learning outcomes (65-95%)	Completion and progression (55-90%)	Learner experience and engagement	Accessibility, flexibility and equity (0-4)	Integrity of assessment	Strategic control / autonomy (20-90%)	External relevance and employability (65-95%)
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AI-Enabled Student Guidance	0.500	0.594	0.000	0.500	0.000	0.500	0.455
Elite Mentor Network / Smart Work-Study	0.500	0.500	0.700	0.000	0.750	0.542	0.688
AI Governance Board / Assessment Integrity	0.344	0.250	0.400	0.300	1.000	0.750	0.409
Aalto-Controlled Core Infrastructure	0.438	0.429	0.400	0.300	1.000	1.000	0.455
Aalto-Certified Micro-Credentials	0.344	0.429	0.200	0.500	0.500	0.583	0.594
Pedagogical Training for Staff	0.600	0.594	0.400	0.300	0.750	0.583	0.500
Nordic-Baltic Tech Alliance	0.750	0.688	0.700	0.300	1.000	0.917	0.850

Appendix F. Portfolio input data and base-case selection

Action	Cost (M€)	Time (years)	Capacity points	Category	Base-case status
AI-Enabled Student Guidance	1.2	1.5	2	Student Support & AI	Not selected
Elite Mentor Network / Smart Work-Study	1.0	1.0	1	Work-Integrated Learning	Selected
AI Governance Board / Assessment Integrity	0.5	0.8	1	Governance & Integrity	Not selected
Aalto-Controlled Core Infrastructure	2.0	2.0	2	Digital Infrastructure	Selected
Aalto-Certified Micro-Credentials	1.1	1.2	1	Flexible Learning Pathways	Selected
Pedagogical Training for Staff	0.8	1.0	2	Faculty Capability	Selected
Nordic-Baltic Tech Alliance	1.4	1.5	1	Partnerships & Reputation	Not selected

Appendix G. Sensitivity-analysis cases and results

Case	p(CGC)	p(ALS)	p(HFU)	Budget (M€)	Capacity	Max actions	Time limit	Optimal utility	Selected actions	Portfolio changed?
Base case	11.1%	61.1%	27.8%	5.0	6	4	10.0	2.46	Elite Mentor Network / Smart Work-Study, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials, Pedagogical Training for Staff	Base
Equal probabilities	33.3%	33.3%	33.3%	5.0	6	4	10.0	2.49	Elite Mentor Network / Smart Work-Study, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials, Pedagogical Training for Staff	No
Scenario 1 favored	50.0%	25.0%	25.0%	5.0	6	4	10.0	2.47	Elite Mentor Network / Smart	No

									Work-Study, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials, Pedagogical Training for Staff	
Scenario 3 favored	20.0%	20.0%	60.0%	5.0	6	4	10.0	2.55	Elite Mentor Network / Smart Work-Study, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials, Pedagogical Training for Staff	No
Budget tightened	11.1%	61.1%	27.8%	4.5	6	4	10.0	2.45	AI-Enabled Student Guidance, Elite Mentor Network / Smart Work-Study, Pedagogical Training for Staff, Nordic-Baltic Tech Alliance	Yes
Budget relaxed	11.1%	61.1%	27.8%	5.5	6	4	10.0	2.49	AI-Enabled Student Guidance, Elite Mentor Network / Smart Work-Study, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials	Yes
Capacity tightened	11.1%	61.1%	27.8%	5.0	5	4	10.0	2.27	Elite Mentor Network / Smart Work-Study, AI Governance Board / Assessment Integrity, Aalto-Controlled Core Infrastructure, Aalto-Certified Micro-Credentials	Yes
Max actions tightened	11.1%	61.1%	27.8%	5.0	6	3	10.0	1.91	AI-Enabled Student Guidance, Elite Mentor Network / Smart Work-Study, Pedagogical Training for Staff	Yes

Appendix H. SWOT tables

SWOT table for Scenario 1:

Strengths • Strong academic reputation and international visibility • High teaching and research standards. • Strong campus community and interdisciplinary culture • Existing digital and innovation capabilities	Weaknesses • Limited capacity to provide highly personalized support for all students • Uneven faculty readiness for AI-supported teaching and assessment • Dependence on external AI and digital platform providers • Risk of lower accessibility if education becomes more selective
Opportunities • Attract top international students and faculty • Strengthen Aalto's position as a high-quality campus-based university • Use AI to support feedback and learning guidance • Build a trusted model for ethical and human-led AI use in education	Threats • Increasing global competition among leading universities • AI use may create fairness issues between students with different technical skills • Over-reliance on external AI providers may weaken academic autonomy • Stronger selectivity may increase pressure, stress and inequality among learners

SWOT table for Scenario 2:

Strengths • Strong capability to develop modular and flexible learning offerings • Existing expertise in digital learning and AI-supported tools • Strong links with industry and professional networks • Interdisciplinary profile suitable for lifelong learning needs	Weaknesses • Limited capacity to support different learner profiles at scale • Risk of fragmented learning paths and weaker degree identity • Uneven readiness among faculty to design modular and personalized learning • Possible dependence on external platforms for scalable digital delivery
Opportunities • Growing demand for lifelong learning and continuous upskilling • More learners need flexible study options alongside work and family life • AI can support personalized guidance, feedback and progress tracking	Threats • Competition from global online universities and edtech providers • Learning may become too automated or impersonal • Algorithmic guidance may reduce human judgment and student agency

• Micro-credentials can attract alumni, professionals and non-traditional learners	• Modular learning may weaken community, academic depth, and long-term engagement
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SWOT table for Scenario 3:

Strengths • Strong academic values and trust in human-led education • Good foundation for mentorship, project work and community-based learning • Credibility in responsible innovation and ethical governance • Strong Nordic and European collaboration potential	Weaknesses • Human-centered teaching can increase staff workload • Limited scalability if AI and automation are used only selectively • Slower adoption of new digital tools may reduce efficiency • More intensive teaching formats may require more resources
Opportunities • Strengthen Aalto's reputation as a trusted and responsible university • Build deeper student-teacher relationships and stronger learning communities • Develop high-quality assessment through oral exams, projects and human judgment • Position Aalto as a leader in ethical and resilient education	Threats • Risk of falling behind universities using AI more aggressively • Higher workload may create pressure on faculty and staff • Human-intensive learning may be costly and difficult to scale • Too cautious AI use may reduce innovation and international competitiveness